

Cheran Senthilkumar

cheran.v.senthil@gmail.com · +1 574-329-2075 · Remote · US Citizen
github.com/cheran-senthil · Codeforces: c1729

Infrastructure & backend engineer · 7+ years · real-time & distributed systems · web3 · algorithmic optimization
Available for remote contract & founding-team engagements

SKILLS

Languages TypeScript / Node.js, Go, Python, C++, React
Infra AWS, GCP, Kubernetes, Docker, Terraform, CI/CD, monitoring & scaling
Data PostgreSQL, Redis
Systems Real-time / WebSocket services, distributed state, web3 / crypto infrastructure, fraud & trust scoring

EXPERIENCE

Independent Software Engineer & Consultant, Self-employed · Remote Apr 2025 – Present

- Built a **real-time multiplayer platform**: a TypeScript monorepo with a low-latency authoritative WebSocket game server, distributed state synchronization, and fault-tolerant session handling.
- Owned the full stack: backend, real-time client, deployment, and infrastructure security; optimized the core engine's algorithms for latency.

Founding Engineer, EarthFast · Remote (US) Apr 2024 – Apr 2025

- First engineering hire at a **decentralized frontend-hosting protocol** (on-chain staked nodes serving dApp frontends; raised **\$1.4M** pre-seed).
- Built backend services in Go and owned the cloud infrastructure as an independent IC.
- Established AWS infrastructure: CI/CD pipelines, monitoring, and scaling.

Senior Software Engineer, TipTop · Remote (US) May 2023 – Jan 2024

- Built core platform APIs and backend services for an instant-cash electronics resale and trade-in marketplace (Postmates' founder, a16z-backed).
- Designed and implemented a **trust-scoring system** for platform-wide fraud detection.
- Set the research direction for two data scientists building fraud-prevention models from user-behavior analysis.

Infrastructure Engineer (early employee), Audius · Remote (US) Dec 2020 – Apr 2023

- Early infrastructure engineer at a crypto unicorn, a decentralized music-streaming network with **5M monthly active users**.
- Owned multi-cloud deployment and orchestration across AWS and GCP; cut **\$20k/month** in cloud costs.
- Built a local development stack that let engineers run the full system locally.

Senior Software Engineer (Contract), ChessLang · Remote Sep 2020 – Nov 2020

- Full-stack features for a chess-education platform; DevOps and deployment automation.

Software Engineer, Torus Labs (Web3Auth) · Singapore Jul 2019 – Aug 2020

- Built components for **decentralized key management** with OAuth / blockchain verification.
- Created security testing frameworks for cryptographic operations; multi-cloud deployment with failover.

Freelance ML Engineer, AI Palette · Remote Feb 2019 – May 2019

- Predictive analytics and NLP models for FMCG market-trend analysis.

PROJECTS & ACHIEVEMENTS

- PyRival**, competitive-programming library, **1,278 GitHub stars**.
- TLE**, Discord bot for competitive programming, **337 stars**.
- run-on-ec2**, Go CLI for rapid script execution on AWS EC2.
- 1st place (USA), Google HashCode 2019** · **IIT-JEE All India Rank 987** · ProjectEuler+ 277/102K · Codeforces c1729.

EDUCATION

B.Tech, Mathematics & Computing · IIT Guwahati 2014 – 2018

- Class Representative (2014–2018); President, MATRIX (2016–2017).
- Research: congruences of partitions & rank differences; a multivariate extension of the Chinese Remainder Theorem.